Tail Analysis Indicates Lost Seals

immediate

Abstract **Complex polygenic traits are typically modeled as the additive sum of many variants of limited effect alongside rare risk alleles of substantial penetrence. This distinction likely reflects a balance between long-range selection pressure and the limited ability of trait-altering mutant alleles to persist over multiple generations. Identification of individuals with extreme trait values as probable carriers of rare high-impact variants has led sequencing strategies to focus on individuals in the tails of the trait distribution. Tail trait values can also result from extreme polygenic burden created by recombination and reassortment. Therefore, to enhance the efficency of sequencing studies, it is imperative to be able to discriminate between these different etiologies.**

One indicatation that rare variants may play a significant role in the tails of the trait distribution can be reflected in the reliability of polygenic risk scores (PRS) in the tails. If highly penetrant rare variants not included in GWAS determine the tails of the trait distribution such individuals will not be assigned appropriate risk scores. In this study, we develop a *polygenic regression-to-mean* (PRM) statistical test that compares the PRS in the tails of the trait distribution to expectation PRS under a polygenic model. Applying this test to hundreds of traits in the UK Biobank reveals that the majority of quantitive traits exhibit significant PRM in one or both tails.

Next we examine whether rare variants in exome studies are enriched in trait tails with significant tail regression and demonstrate that such tails are more likely to be under negative selection. Further examining these trait tails we find an association between PRM and other measures of fitness that may be influenced by rare variants including the lifetime number of illnesses and still births. We then compare these results to a simulation and find that they match.

Finally, we assess whether the predictions of a complex trait architecture derived from a conditional sibling model align with the observed diversity in the architecture of trait extremes. The concordance between these measures underscores the need to consider the role of rare variants in the analysis of trait tails. The PRS and sibling analyses presented here offer valuable tools for the targeted selection of traits and samples likely to harbor de novo or rare deleterious alleles and represent an approach that has the potential to significantly enhance the effectiveness and design of sequencing studies.

Introduction

Genome-wide association studies (GWAS) are instrumental in understanding the genetic architecture of complex traits. These studies typically identify numerous common genetic variants, each of small individual effect (1) that can be used in summation to make predictions. The idea that these common variants adequately represent the genetic architecture is supported by evolutionary theory, which suggests that many human traits (e.g., height, birth weight) are under stabilizing selection, which reduces the fitness impact of large-effect variants (2).

A recent hope for advancing personalized medicine is in the potential of polygenic risk scores to identify high-risk individuals who may benefit from early interventions, such as adoption of specific lifestyle regimes or increased screening (3–6). However, this strategy will be underpowered when applied to traits or diseases with reduced PRS in the tail of the distribution due to reduced common polygenicity. To overcome this problem, the genetic aetiology underlying the tails of phenotype distributions must be investigated and better characterized (7). Polygenic prediction models use these common small-effect variants to generate polygenic risk scores (PRS) whose performance making out-of-sample predictions is usually evaluated with aggregate estimates like *variance explained* (R^2). However, new evidence has suggested that this level of analysis may not be adequate because the preformance of prediction models is not uniform across the trait distribution. This

non-uniformity results because the contribution of small effect common variants relative to highly penetrant rare variants and the environment is not equal across the trait distribution (7, 8).

This negative correlation between effect size and minor allele frequency reflects the interplay between long term selection against extreme trait values and the ability of mutant alleles with significant effects to escape this trend through chance or short term environmental changes. Recent or ongoing selection can leave distinct genetic signatures, such as an enrichment of rare variants in the tails of trait distributions during directional or stabilizing selection. The expectation that individuals with extreme trait values are more likely to carry rare large-effect variants has led to strategies that priorize sequencing of individuals in the tails of the trait distribution (9, 10). However, tail trait values can also result from extreme polygenic burden created by recombination and reassortment, and a more efficient approach would involve considering more lines of individuals to allow the selection of individuals with strongest evidence of rare variant architecture for sequencing.

Here we attempt to distinguish whether the tails of the distribution of different traits are primarily driven by common polygenicity or by a more "monogenic-type" architecture. We accomplish this by developing a statistical test that measures the fit of the polygenic model across the trait-PRS distribution and identifies non-polygencity in each tail and applying it to over a thousand UK Biobank traits (see ***Figure 1***). Next we consider whether evidence of negative exists in a subset of uncorrelated traits and show that the trait tails under negative selection are more likely to be associated with rare variants. We further triangulate evidence for selective constraint leading to reduced common polygenicity, by testing for the fertility of individuals in the tails based on their known measures of fitness that correlate with de novo mutations (11, 12).
Finally, we assess whether the predictions of a complex trait architecture derived from a conditional sibling model align with the observed diversity in the architecture of trait extremes. The striking concordance between these measures underscores the need to consider the role of rare variants in the analysis of trait tails. The PRS and sibling analyses presented here offer valuable tools for the targeted selection of traits and samples likely to harbor de novo or rare deleterious alleles and represent an approach that has the potential to significantly enhance the effectiveness and design of sequencing studies.

We use PRS to assess the contribution of rare large-effect variants on extreme trait values. Rare variants are typically not included in PRS, leading to reduced accuracy of the PRS for individuals with rare large-effect variants. This inaccuracy will be most apparent in the tails of the trait distribution where individuals with large-effect variants are most likely to reside. Specifically, PRS values will tend to converge toward the mean PRS in the tails – *PRS-regression to the mean*.

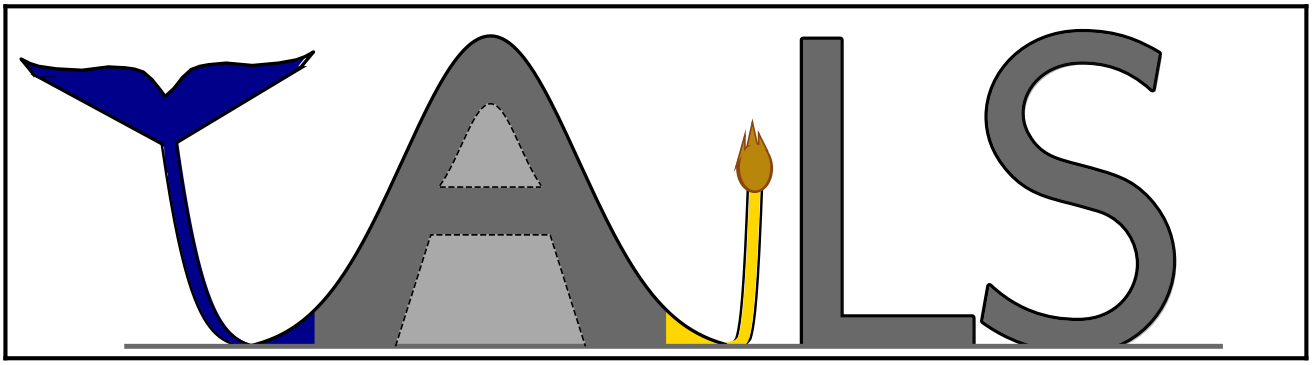

Fig. 1: **Placeholder for Schematic**

Results

The primary goal of this study is to identify and characterize *polygenic-regression to the mean* (PRM), which occurs when individuals in the tails of the quantitive trait distribution exhibit polygenic risk that is closer to the mean risk score than expected. To quantify this phenomenon we developed a quantitive trait statistical test (PRM-test - see methods) that estimates effect size, direction, and signficance of movement away from the expected polygenic load for individuals in the lower and upper 1% of the population. Applying this test to the UKB reveals that 400 of 1,185 heritiable ($h^2 > 0.01$) quantitive traits have signficant ($p < 0.05$) movement from expectation in one or both tails and that 95% of the time this movement is toward the mean (see - Supplement). Observing that many of these tests are underpowered due to insufficient sample size and poor polygenic prediction (R^2), we identify a set of 75 high-powered ($h^2 > 0.05$,$R^2 > 0.01$) non-redundant, quality-controlled traits for further analysis (see Methods - QC). Using these 75 traits we perform PRM-Testing, rare-variant analysis, analysis of selection to help elucidate the mechanism that may be behind PRM. We then replicate these results using a sibling model and different datasets.

Polygenic Regression To The Mean (PRM) is Common

Running our PRM-Test on our 75 QC'ed traits reveals that the 80% of tails (121 of 150, FDR level 0.05) exhibit movement away from expectation and that 98% (118 of 121) of the time the movement that occurs is regression toward the mean. The majority of traits (47 of 75) were classified as "Dual Tail" (significant regression to the mean in both tails), while 15 were classified as "Lower Tail" and 9 were classified as "Upper Tail" (see ***Figure 2b***). For illustrative purposes we identified four "index traits" to represent "No Regression" (Triglycerides - the highest h^2 of any trait without tail regression), "Lower Tail" (Haemoglobin Concentration - most signficant), "Upper Tail" (RBC Distribution Width - most significant), and "Dual Tail" (Sitting Height - most significant maximum pvalue). The tail dynamics observed in each of these traits can be illustrated in ***Figure 2a***.

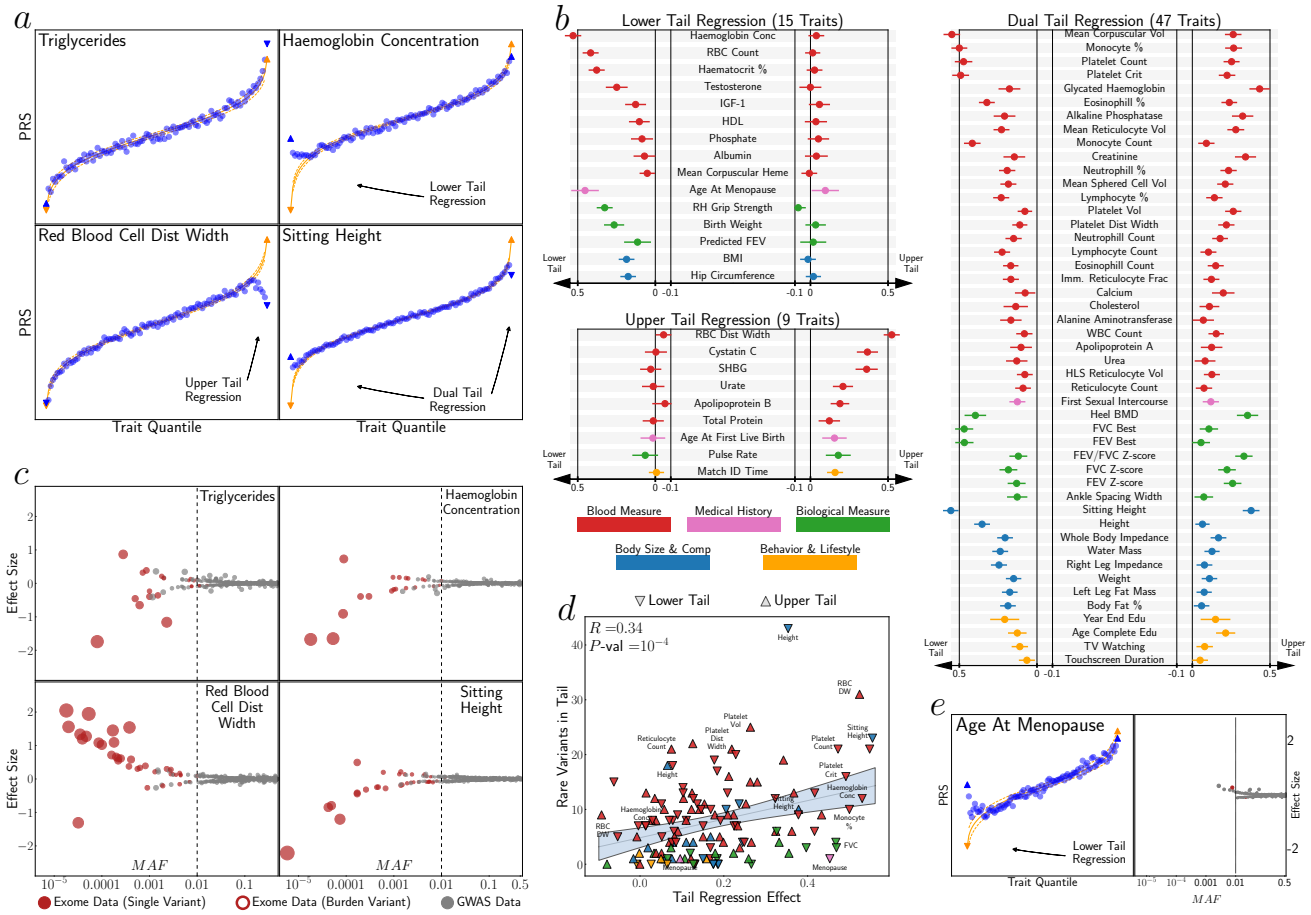

Fig. 2: **(a)** Four traits the demonstrate PRS tail regression (clockwise: No Tail Regression, Upper Tail, Lower Tail, Dual Tail). **(b)** Miami plots demonstrates that dual tail regression is dominant tail regression observed. **(c)** Trumpet plot reveal highly penetrant rare variants in many of the trait tails where PRM is obersved. **(d)** Tail regression is associated with greater a number of penentrant rare variants. **(e)** Age of Menopause is a counterexample.

The Role of Rare Variants in PRM

The exclusion of rare variants from PRS (traditionally a cutoff of minor allele frequency (MAF)>1%) has been shown to reduce model performance (13). To investigate whether PRM is a consequence of rare variants of large effect that drive individuals into the tails of the trait distribution we created trumpet plots (***Figure 2***c) to visualize the impact and direction of of rare variants from GWAS (14) and exome studies (15). Focusing on our single tail index traits, we notice that there are a greater number of rare variants in the lower tail of *Haemoglobin Concentration* and in the upper tail for *Red Blood Cell Distribution Width*. Additionally, for *Sitting Height* there are more rare variants in the lower tail (where the PRM Effect is greatest) and in *Triglycerides* there are also more rare variants in the lower tail (where PRM Effect is greatest, albiet non-signficant). This result is consistent with the hypothesis that rare variants of large effect that not accounted for in the common-variant PRS responsible for the tail specific PRM. In ***Figure 2***d we observe a small but signficiant relationship between the magnitude of regression and the number rare variants in the tail. However, for many traits the tail with more PRM shows almost no rare variants (see ***Figure 2***e) suggesting that strong selection may result in private variants that cannot be identified using population based sequencing.

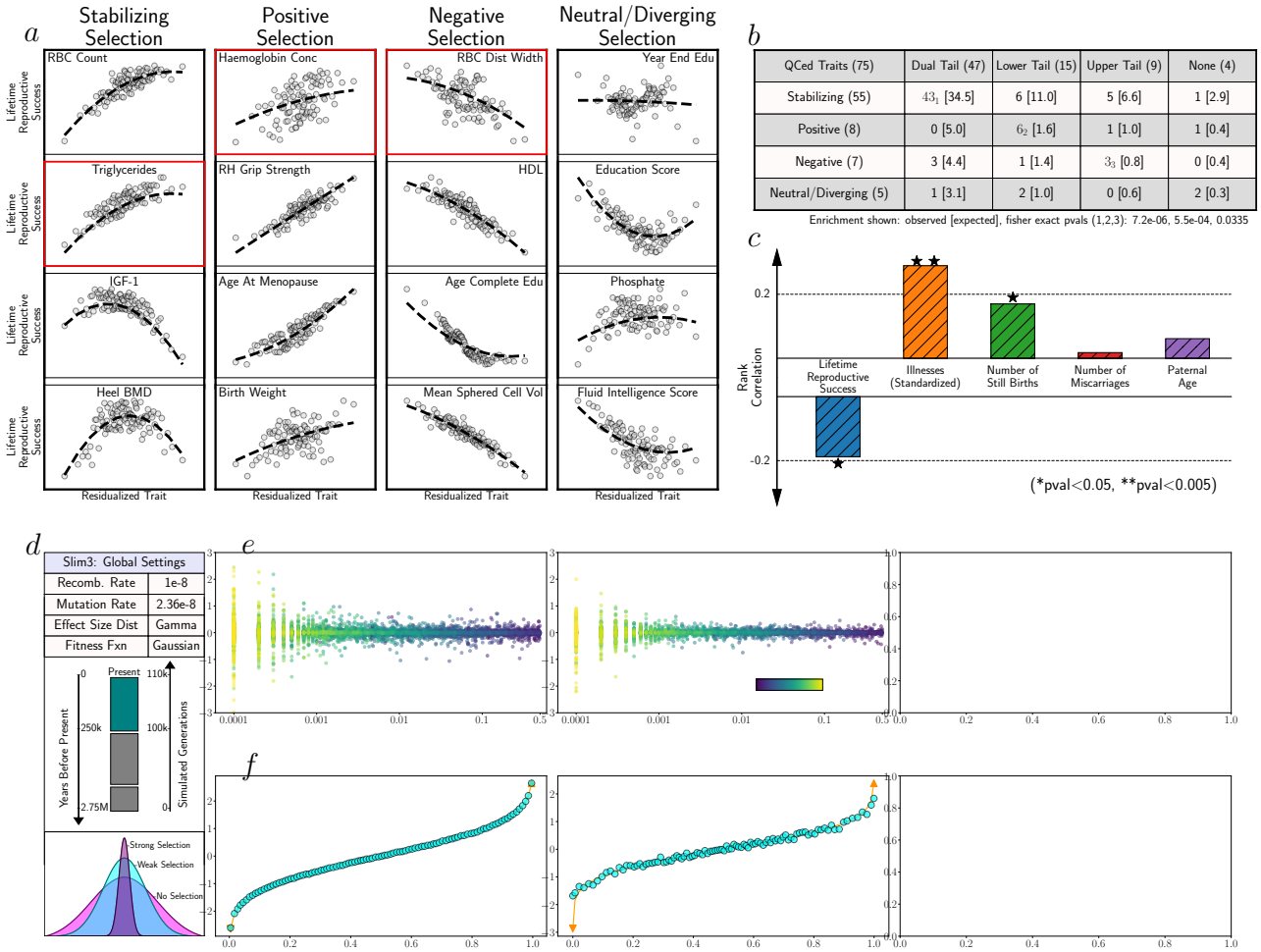

Fig. 3: **(a)** Relative Lifetime Reproductive Success used to approximate trait selection. **(b)** Association between negative tail selection and PRS regression. **(c)** Tail Fitness Association. **(d)** Simulaton by Anil. **(e)** Simulation by Anil.

The Impact of Selection on PRM

Using adjusted lifetime reproductive success (2, 16) as a proxy for selection we classify (see Methods - LRS) each trait with respect to whether it is under *Stabilizing, Positive, Negative, or Neutral/Disruptive* to produce results in agreement with a more comprehensive sex-specific analysis (See ***Figure 8***). The majority of traits analyzed here (55 of 75) are classified as under stabilizing selection which is characterized by lower fecundity in both tails of the trait distribution. Stabilizing selection may be the result of general fitness favoring the trait mean or the result of sex specific selection that averages to produce this relationship (2). Eight traits were classified as being under positive selection and seven traits were classified as

being under negative selection. We also classified four traits as under neutral selection (no signficant relationship with LRS) and one trait as divergent/disruptive selection where fecundity is higher in both tails that at the mean of the distribution (***Figure 3***a). We find a significant relationship between selection classification and PRM classifition (***Figure 3***b), whereby traits under stabilizing selection are enriched in "Dual Tail Regression" ($p < 2.5 \times 10^{-5}$), traits under positive selection are enriched in "Lower Tail Regression ($p < 2.4 \times 10^{-4}$), and traits under negative selection are enriched in "Upper Tail Regression ($p < 0.024$). Focusing on each tail separately we find an even stronger relationship insofar as the in the 118 trait tails which exhibit PRM, the tail is selected against 110 times (93%) while in the 32 tails which do not exhibit PRM the tail is only selected against 16 times (50%) ($pv < 0.001$).

Tail Fitness

In addition to *lifetime reproductive success*, we analyzed four measures of fitness that are likely to be associated with reduced fitness and rare variant or *de novo* etiology. Here, we note that under the hypothesis that rare variants under negative selection are responsible for PRM, we predict that trait tails with large effect PRM will have lower average *lifetime reproductive success* and higher average numbers of *Still Births* and *Miscarriages*. Additionally, traits tails with large effect PRM are predicted to have more standarized illnesses because both result from large effect variants, although it is possible that downstream effects of illnesses results in extreme trait values. Finally, if paternal age is associated with more *de novo* mutations (26) it should also be associated with PRM.

In ***Figure 3***c the rank correlation for the average tail fitness measure and the PRM effect size across the trait tails is shown. We observe that the direction of rank association is in line with the rare-variant hypothesis in every case, although the association between PRM and the number of miscarriages and paternal age are not significantly associated.

Simulating Selection (Anil's Result)

In ***Figure 3***d we confirm this result using a simulation (see *Methods*) where we simulate many generations of stablilzing selection, after which the ratio of contribution to heritability by rare variants goes up and PRM is observed.

Replication in Siblings

Discordance between siblings in the tails of the trait distribution is indictive non-polygenic inheritance (17). The conditional sibling distribution under polygenicity can be used to both to estimate heritability and to estimate the impact that rare variants have on tail genetic architecture (18). In ***Figure 4***a the expected impact of polygenicity, de novo, and mendelian rare variants are illustrated.

Applying this model to the sibling data in the UKB to estimate h^2, we find a correlation between snp h2 and sibling h2 ***Figure 4***b. In our index traits we use this model to infer the presence of rare variant architecture in the tails and find that lower tail discordance in *Haemoglobin Concentration*, upper tail discordance in *Red Blood Cell Distribution Width* and dual ail discordance in *Sitting Height*, all of which match the result from our PRM test. Overall we find a significant relationship between our sibling test pvalue and the pvalue from our PRM-test ***Figure 4***d.

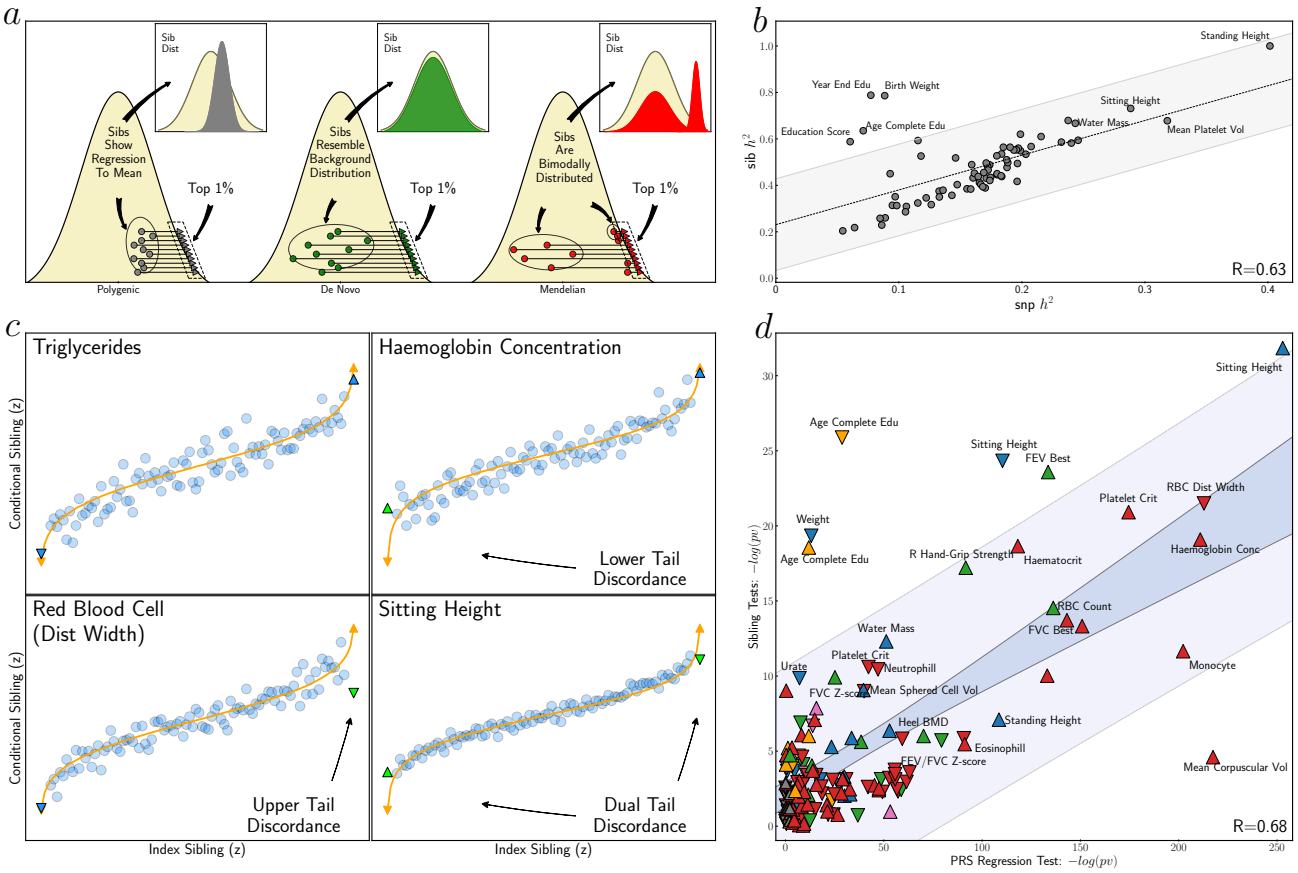

Fig. 4: **(a)** Sibling Model **(b)** Association between SNP heritability and Sibling Heritability **(c)** Sibling Concordance Tests **(d)** Relationship Between Sibling Test and PRM-Test

Replication

We performed three *flavours* of replication.

1. **Repeated Subsets of The Target Data:** Assess the reliability of our PRM-test in smaller samples
2. **Non-European Ancestry Inviduals in UKB:** PRM replication across ancestry (within biobank)
3. **European Ancestry Individuals in All Of Us:** PRM replication across biobank (within ancestry)

Bootstrap Analysis using Target Subsets

To assess the performance of our PRM-test using smaller samples we re-sampled small subsets of our target populations, and performed our PRM-test on these subsets. In 5d we notice that even the correlation of effect sizes was approximately 0.70 even when the subset size was only 5k, however concordance in discovery was poor at these levels.

Out of Sample Replication

The correlation between PRM effect-sizes was greater in the UKB Non-European Ancestry data despite smaller sample sizes, although the number of true positives identified was lower (underpowered). This suggests that even if PRS isn't portable across ancestry, that regression to the mean is portable and in fact may be even more than portable.

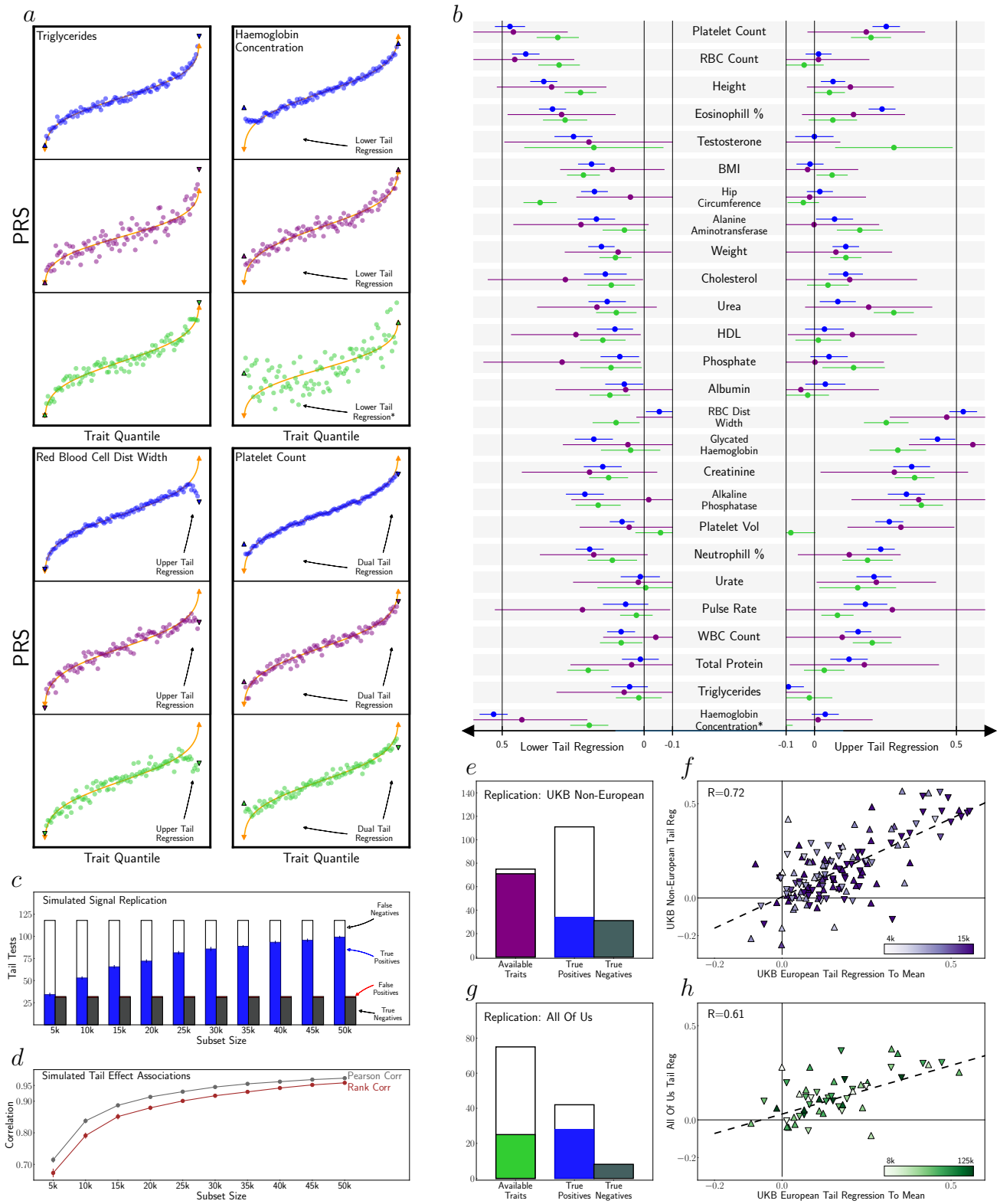

Fig. 5: **(a)** Four Index Traits are replicated In UKB Non-European Cohort and the All Of Us Cohort. **(b)** Miami plots demonstrating replication across 25 Traits available in UKB-Non-European and All Of Us Cohort, plus hemoglobin **(c)** Simulated Signal Replication: Increasing larger subsets were sampled and the replicability relative to our UKB result was calculated **(d)** Same thing but correlation **(e,f)** Replication In Non-European **(g,h)** Replication In All Of Us

Discussion

In this study, we propose the use of PRS Regression-to-the-Mean (RM) Test as a way to identify traits that are enriched for rare or de novo – more "monogenic-type" effects (MTE) in extreme. We applied this method to a list of quantitative traits available in UKB. In addition to genotype data, we also incorporated the rich source of UKB information including the sibling data (Sibling RM Test), and paternal age (Paternal Age Test) and number of children (Fecundity Test) into our analyses. We found directional consistency in most of the tests performed, providing converging evidence for the feasibility of using PRS as a strategy to identify traits with potential MTE. In our analyses, the PRS RM test can be used to identify both the "polygenic" and "MTE" event, we hypothesize that for a highly polygenic trait, samples with extreme trait values (ETV) are expected to have extreme PRS correspondingly. Deviation from the expectation may suggest samples at the extreme tails are enriched for MTE that are under selection. Our results show significant correlation in the ranking traits from least to most evidence of common polygenicity in the tails of their distributions between the PRS RM test and Fecundity test. Furthermore, the results of the PRS RM test are supported by the Sibling RM test of which we use the sibling phenotype data to investigate the common polygenicity in different traits. Sib-pairs are expected to have similar trait value under the influence of common polygenicity. On the other hand, if the ETV in sib A is contributed by de novo mutations, we expect sib B to be discordant phenotypically. An alternative explanation for discordant phenotype in sib-pairs could be due to the non-shared environment factors with large effect.

One major limitation of these tests is that they are highly sensitive to measurement errors. Measurement errors would result in the same PRS/ sibling departure at the extreme ends, because the PRS is no longer reflective by one's phenotypic measurement. In addition, both the PRS RM and sibling RM Test cannot distinguish between large effect mediated by environmental factors and MTE. For example, if a person with extremely low levels of LDL but normal PRS can be caused by malnutrition. Therefore, collection of the associated environmental factors and the corresponding adjustment becomes vital to alleviate this problem when applying these tests. Moreover, these tests are only applicable to quantitative traits, more work is needed to develop for testing binary outcome. In this study, we identified traits with evidence to be influenced by MTE at the trait tail(s), however, we did not identify or confirm the corresponding de novo mutation. Despite the lack of parent-offspring trios which is essential for the identification of de novo mutations, it is technically challenging to define a list of candidate genes and variants for every trait we investigated. Further investigation is warranted to validate this finding. In conclusion, we propose a novel way of using polygenic risk score to look for evidence of de novo/ rare mutations. This can be used as a method to identify traits that are more likely influenced by de novo mutations, and select samples who are more likely carry de novo mutations, potentially maximizing the power of a sequencing study. This method can be applied to any traits with continuous measure such as age of onset, disease severity and rate of disease progression.

Methods

Data Processing

UKB genetic data

Issues:

- More specifics on the numbers.
- Explain ancestry clustering more specifically (which ancestries?
- What PC values chosen?). Nobody could repeat this accurately as written.
- Some loose language in places. Several places where citations needed e.g. "recommended by the UK Biobank team." i
- Need some better context - e.g. In the primary analyses in unrelated individuals (Figures 2, 3, 5)...
- In the analyses performed on siblings (Figure 4)...
- Not sure if Mixed ancestry results needed anymore?

The UK Biobank is a prospective cohort study that included approximately 500,000 participants with measurements of 4,000 phenotypes, recruited across the United Kingdom from 2006 to 2010 (19). The genetic dataset consists of 488,377 samples genotyped at 805,426 SNPs. To determine population ancestries, 4-means clustering analysis was conducted on the first two Principal Components (PCs) of the genotype data. Each cluster's ancestry was determined based on the majority of individuals' country of birth (field ID: 20115) within that cluster.

Subsequent to clustering, standard quality control (QC) procedures were applied independently to each ancestry cluster. SNPs with a minor allele frequency < 0.01, genotype missingness > 0.02, or Hardy-Weinberg equilibrium test P-value $< 10^{-8}$ were excluded. Samples exhibiting high levels of missingness or heterozygosity, inconsistencies in genetic-inferred and self-reported sex, or displaying aneuploidy of the sex chromosomes were removed in accordance with recommendations from the UKB data processing team (**?**). A greedy algorithm was employed to remove related individuals that maximizes samples retention while removing all third-degree relatives (kinship coefficient > 0.044) (20). After this QC there 432k unrelated individuals remained. Primary analyses were performed on over 410k unrelated European samples and replicated was performed using the 22k samples classified as non-European ancestry.

For the sibling based analyses sibling pairs were selected by first identifying first degree relatives defined as those with kinship coefficient within the range 0.177 and 0.354, this range either side of the expected value of 0.25 was used to account for the expected variation in within-pair similarity (22). The proportion of SNPs with zero identity-by-state (IBS0) was used to distinguish sibling pairs from parent-offspring pairs since parent–offspring pairs have IBS0≈0. It has been shown in the UKB there is good separation between pairs of first degree relatives at IBS0=0.00112(22) thus sibling pairs were selected as those that exceeded this threshold, this gave a total of 17,869 sibling pairs (17,289 of European ancestry, that were used in our primary analysis).

Quantiative Trait QC

Issues:

1. h2 > 0.02 - why?
2. Seems a bit unusual to describe the QC, then the residualisation/normalisation, and then some further QC steps on the untransformed data. Is this the best order?
3. I think the reduction from 151 to 92 traits due to rg<0.99 cut-off is likely to raise alarm bells - how many more would've been lost if you'd used 0.98, 0.95, 0.9 instead?
4. There are a lot of P-values reported in the paper dependent on a test of many traits.. but if those are highly correlated then the overall P-value reported will be inflated. We've talked about this before I know but isn't this still a problem!?

Solutions:

1. Typo - 0.01, still arbitrary, but better.
2. I don't know. But basically Clives PRM-QC finds non-linearities but we know our normalization can hide them, so it does make sense to be conservative here.
3. Filtering on Clives QC first and doing genetically correlations last (as we should), it hardly makes a difference, I changed it to 0.95 and gave some numbers that should calm alarm bells.
4. They are not highly correlated - but I think we have done enough in the methods to make people calm.
5. This does need to be talked about in the discussion - the whole point of the paper is TAILS, and the tails are nothing like the main part of the distribution, so highly correlated body doesn't mean much about the tails.

Quantiative traits were selected in UKB as those with at least 10 distinct values, heritability $h^2 > 0.01$ (measured by LD score (23) applied to publicly available UKB GWAS summary statistics (14)) and >10,000 observation in unrelated samples, this gave 1,158 traits. Traits values more than six standard deviations from the mean were removed. Traits were then standardised as follows: (1) linear regression models were applied to each trait, with covariates for age, sex, recruitment centre, genotyping batches, and the first 40 genetic principal components, (2) standardised values were obtained by inverse normal transforming the rank of the residuals.

Results are presented in the Supplementary Material for the PRS-regression test on all 1,158 continuous traits, 1,103 of which were sufficiently linear after transformation to produce an estimate of effect size. Our analyses assume that the transformed trait values are Gaussian distributed with primarily additive polygenic architecture. Therefore the 1,103 traits that passed our PRM-test QC were further thinned by enforcing the following criteria. **Continuity:** top two modal untransformed trait values include less than half of samples. **Unimodal:** Dip-Test ((24)) P-value > 0.05 applied to untransformed trait values. **Non-skewed:** skew of untransformed trait values < 2. **Polygenic:** Maxmimum single variant $r^2 < 0.5\%$. Finally, to ensure a minimum power threshold, traits with PRS $r^2 < 1\%$ or fewer than 50,000 samples in the target data (after filtering outliers) were removed. This resulted in 122 traits. To ensure a non-redundant survey of traits, pairwise genetic correlation were calculated using (23) and 60 traits were identified as gentically unique (no pair w $r_g > 95\%$, average top match = 0.60, standard-deviation = 0.2) and kept in the analysis. Additionally, 14 traits has a single almost identical match (eg: "Standing Height" (ID:50) and "Height Measured Prior to Imaging" (ID:12144)), in these cases the trait with greater sample size was priotized. The other 48 traits were part of larger clusters of genetically similar traits, the largest of which related to measurements related to of body-mass (eg: body fat percentage, leg mass, arm mass, etc) and heel bone mineral density. Using the same data maximization algorithm applied to extra unrelated individuals (20), we were able to select 9 of these traits which resulted in a set of 75 genetically non-redudant traits.

All of US

All Of Us in biobank that contains XX binary and XX continuous traits from individuals of multiple ancestries. Trait matches were found with an initial database search and then were inspected by eye to determine if trait descriptions were sufficiently similar. This process yielded 32 traits, 25 that passed our PRM-QC. These traits had between 8595 and 127677 total samples (average 73k) in the European Ancestry cohort. For these 50 tails the replication accuracy and replication association was calculated as described above.

Genome-wide association analyses and PRS estimation European UKB samples were split randomly into 50% base and 50% target data sets. GWASs were applied to the base data using PLINK-1.9 (6) applying a linear additive model to the standardised trait values. PRScice-2(v2.3.4) (25) was applied to the GWAS summary statistics of each trait with high-resolution scoring to select the most predictive P-value threshold **NEED: using the target data**. Individual risk scores were calculated for UKB samples in the target data.

Data Analysis

The PRS-regression to the mean (PRM) test

The PRM-test first applies a regression of PRS on standardised trait values. For each trait, the residuals from the regression are then split into quantiles according to the trait distribution and t-tests are applied to test whether the residuals are: (1) significantly less than zero for individuals with trait values in the upper tail and (2) significantly greater than zero for individuals with trait values in the lower tail using t-tests. **NEED: are we now applying two-sided tests?** The tests were applied to individiuals in the top and bottom 1% of the trait distribution.

To avoid false positive associations arising from unusual trait distributions, the residuals of samples in the middle of the trait distribution were tested for departure from the null of zero mean. Specifically, for each trait, two-sided t-tests for zero mean were applied to the residuals assigned in each of the middle 80 centiles of the trait distribution. Traits were deemed to fail QC if the smallest P-value was significant at a Bonferroni-corrected level of 0.05 ($P \leq 0.05/80$). A robust empirical P-value for deviation in trait tails was calculated as follows: for each centile of trait one-sided t-tests for residuals less than and greater than zero were calculated. The P-value for regression to the mean in the upper tail was calculated from the rank of the tests for residuals significantly less than zero and the lower tail empirical P-value from the greater than zero tests. The tests were performed on all 1,121 continuous traits in UKB.

Rare Variant Analysis: Trumpet plots

Trumpet plots show the effect sizes and MAF of independent genome-wide significant variants. Common variants (MAF>1%) were selected as the most significant variant ($P \leq 5 \times 10^{-8}$) in 1Mb windows from publicly available GWAS summary statistics generated using UKB samples of European ancestry (14). Exome sequencing associations from the analysis of UKB European ancestry samples were downloaded (15). Results are plotted for the most significant deleterious, missense or predicted loss of function variants in each gene ($P \leq 2.18 \times 10^{-11}$, one association per gene). Genome-wide significant burden associations are shown where no single variant within the gene is genome-wide significant.

Selection Analysis: Relative Lifetime Reproductive Success

The number of offspring is recorded in the UKB for both females and males, fields *2734* and *2405* respectively. We followed the methodology of Beauchamp (16) and later applied by Sanjak et al (2) which involves splitting samples into four birth cohorts (1934-42, 1943-48, 1949-55, 1956-65) and dividing the number of offspring of each individual by the cohort mean to calculate relative lifetime reproductive success (Y_{rLRS}). The pressence of selective pressure acting on the traits is assessed by fitting the following regression model to relative lifetime reproductive success (2)

$$Y_{rLRS} = \beta_0 + \beta z + \gamma z^2 \quad \textbf{(1)}$$

where z denotes the residualized and normalized trait values and β and γ are linear and quadratic regression parameters. The pressence and type of selection is inferred from the regression parametrers as folows:

- If the P-value associated with the quadratic parameter γ is significant (FDR 5%), then:
  - If $\gamma < 0$, the trait is classified as under stabilizing selection. (i.e. negative selection both tails).
  - If $\gamma > 0$, the trait is classified as under disruptive selection (i.e. negative selection is not occuring in either tail).
- If only the p-value associated with the linear parameter β is signficant (FDR 5%), then:
  - If $\beta < 0$, trait is classified as under negative directional selection (i.e. negative selection in the upper tail).
  - If $\beta > 0$, trait is classified as under positive directional selection (i.e. negative selection in the lower tail).
- If neither parameter is nominally significant, trait is classified as under neutral selection (i.e negative selection is not occuring in either tail.

Selection Analysis: Measuring Fitness in the Tails

In addition to lifetime reproductive success, we consider four additional measures of fitness that are likely associated with negative selection and rare variants:

- **Adjusted number of illnesses:** The number of self-reported non-cancer illnesses (UKB trait Field 2002) is standardized by applying a regression model with covariates for age, sex, recruitment centre and obtaining residuals.
- **Number of miscarriages:** Trait Field 3829.
- **Number of still-births:** Trait Field 3839.
- **Paternal Age:** Paricipant age (Field 21022) is substracted from father's Age (Field 2946) to give the paternal age at birth.

We test whether decreased fitness, as defined by the above proxies of fitness, are assocaited with increased PRM as follows. For each trait, the mean value of the four proxies of selection are calculated for individuals in the upper and lower 1% of the trait distribution separately. Spearman's rank correlation between the mean values for each proxy and the PRM effect size is then calculated across all trait tails.

Selection Analysis: Simulating Stabilizing Selection using SLiM

Using SLiM v4.0 (27), we simulated a quantitative polygenic trait under a standard Wright-Fisher model of stabilizing selection. Stabilizing selection assumes the optimal phenotype is at the mean of the population and that fitness decreases as individual's phenotype moves away from that optimum. Briefly, we first established a population of N=10,000 individuals and simulated a 10N generation long burn-in period to reach an equilibrium state of mutation–drift balance (28).

After neutral genetic diversity was achieved, stabilizing selection on the phenotype was introduced for 1N generations. We used a mutational target size of 1Mb with a constant mutation of $2.36 * 10 - 8$ per base pair (bp) and recombination rate of $1.00 * 10 - 8$ per bp. To simulate new mutations across the mutational target, we implemented a gamma distribution. This is similar to other models of genetic variation in humans that assume a gamma distribution (29), reflecting a long-tail distribution that combines mostly small effects with a small number of relatively large effects. The gamma was parameterized with mean and shape equal to 0.01026 and 0.186. Each mutation that was drawn was multiplied by δ where ($\delta \in -1, 1$) with equal probability, to reflect a symmetrical effect size distribution commonly seen with complex traits. Mutation effect sizes were furthermore multiplied by 1x, 5x, or 10x, to simulate different ranges of mutation effect size. Trait heritability was set to 1.0 which means that the phenotypic variance in the population is fully determined by genetic effects. At each generation, an individual's phenotype (i.e., polygenic score) was defined by the sum of the number of alleles of each mutation, weighted by its effect size.

After the burn-in period, stabilizing selection was implemented through a gaussian fitness function,

$$f_t(x) = baseline + \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\} \times factor \quad \textbf{(2)}$$

where x is the phenotype of the individual at a specific generation t, μ is the phenotypic mean of the population at t and σ the phenotypic standard deviation at the start of selection (i.e., $t = 10N$). The baseline and factor parameters were set at 1 unless otherwise specified. The fitness of each individual was calculated and updated at the end of each generation. Stabilizing selection was simulated until generation 11N after which the simulation was completed.

Simulation data was collected by logging, at each generation, the phenotypic mean and standard deviation of the population, statistics of mutations present in the populations, including allele frequencies and effect sizes, and the mutational profile of each individual. To investigate the contribute of low-frequency variants, a second genetic value was calculate using only variants with $MAF > 1\%$.

Sibling Concordance: Assessing the contribution of rare large-effect variants on extreme trait values

The conditional distribution under polygenicity of an individual's trait value given their siblings trait value is predicted by theory (18)

$$p(s_2 \mid s_1) \sim \mathcal{N}\left(\frac{1}{2}s_1 h^2, 1 - \frac{h^4}{4}\right) \tag{3}$$

where s_1 and s_2 denote the index and conditional-sibling trait values respectively and h^2 is the trait heritability. Rare large-effect variants can lead to deviations from this distribution, particularly in the tails. Specifically, rare large-effect variants within families, *Mendelian effects*, will result in excess concordance between siblings when they both inherit the effect variant. Conversely, a large effect *de novo* mutation in one sibling will result in excess discordance. The sibling test for *Mendelian* effects applies a binomial test for a greater than expected number of sibling pairs both lying in a defined quantile of the tail of the trait distribution by calculating the following z-statistic

$$z = \frac{r - n\pi_0}{\sqrt{n\pi_0(1-\pi_0)}}$$

where n is the number of index siblings in the quantile q under test, r is the number of siblings pairs both in q and π_0 is the expected proportion of sibling-pairs in the quantile under polygenicity

$$\pi_0 = \frac{1}{n}\sum_{i \in A_q}\left(1 - \Phi\left(\frac{\Phi^{-1}(q) - \frac{s_1^{(i)}h^2}{2}}{\sqrt{1-\frac{h^4}{4}}}\right)\right)$$

where A_q is the set of index siblings in trait quantile q, Φ is the normal normal cumulative density function, Φ^{-1} is its inverse and n is the number of sibling pairs.

The sibling test for *de novo* effects constructs a z-statistic for the distance between index siblings in the tail of the trait distribution and their conditional siblings as follows

$$z = \frac{\sum_{i \in A_q} s_2^{(i)} - \frac{1}{2}s_1^{(i)}h^2}{\sqrt{n(1-\frac{h^4}{4})}}$$

For index siblings in the lower and upper tails the alternative hypotheses of $H_1 : z > 0$ and $H_1 : z < 0$ are tested respectively. For further details see Souaiaia et al (18). Heritability h^2 required for the *Mendelian* and *de novo* tests is estimated from the sibling pairs by taking the maximum likelihood of h^2 under the null hypotheses of both tests

$$\log \mathcal{L} \propto -n \log\left(1 - \frac{h^4}{4}\right) - \frac{1}{1-\frac{h^4}{4}}\sum_{i=1}^{n}\left(s_2^{(i)} - \frac{s_1^{(i)}h^2}{2}\right)^2$$

For each trait P-values from the *Mendelian* and *de novo* tests are combined using Fisher's method to give a χ^2 statistic with 4 degrees of freedom,

$$\chi_4^2 = -2(\log p_{Mendelian} + \log p_{denovo})$$

that can be used to generate a meta-analysis p-value to be compared to the PRM-Test.

Replication

Bootstrap Analysis using Target Subsets

Our 75 QC-validated traits were required to have at least 100k total samples. In the target sample (half the total), after outlier removal our QC-validated traits had between of 50,085 and 185,362 (average 142k) target samples. To explore whether our PRM-test results replicate with smaller sample sizes, we repeatedly sampled subsets of between 5k and 50k target samples and performed our PRM-test on these subsets. For each value between 5k and 50k (counting by 5k) we performed 100 trials across each of the 150 trait tails and calculated the **(1)** relative accuracy (true and false positives and negatives) relative to our full dataset (FDR 5%) and **(2)** the linear and the rank correlation of for the regression effect size across all the tails.

Replication: Non-European Ancestry Individuals in the UKB

Our Non-European Ancestry samples in the UKB have between 3,150 and 13,037 (average 10k) total samples for our 75 QC-validated traits. Applying our PRM-test using the European Ancestry PRS to the Non-European target, 71 traits passed our PRM-QC. For these 142 tails we calculated the replication accuracy and replication association as described above.

NEED: ***From Beatrice*** Genotype and phenotype QC.

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Supplement

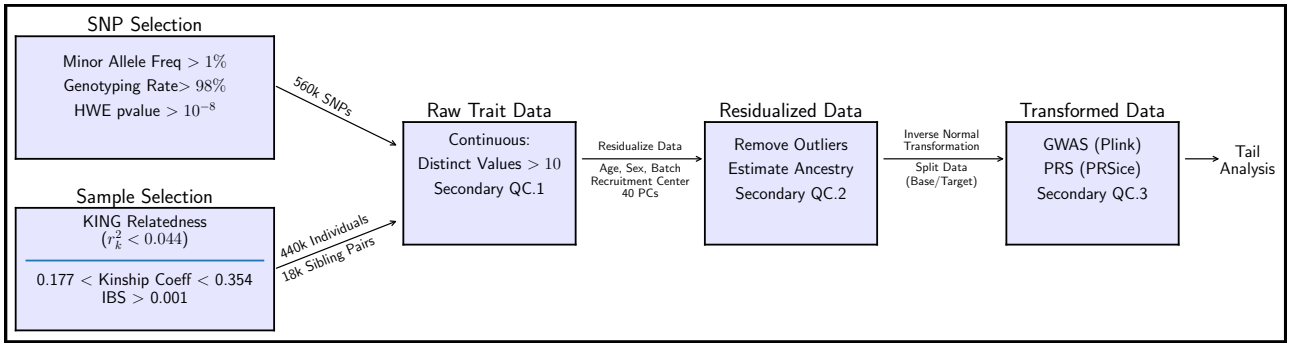

Fig. 6: **Sample and SNP Workflow**

Filter	Classification	Criteria	Where Applied	Result [Running]
Unique	Unique/Redundant	No Genetic Duplicates ($r_g < 99\%$)	Inverse Transfomed Data	(133,131)
Continuity	Continuous/Semi-Cont	Top Two Model Values < 50% of data	Raw Trait Data	(247,17), [122,11]
Normality	Normal/Alternative	Unimodal (Dip Test > 5%), Symetric ($\gamma < 1.5$)	Residualized Data	(202,62), [92,30]
Additivity	Polygenic/Oligogenic	max(Variant r^2) < 5%	GWAS Data	(261,3), [91,1]
Heritability	Heritible/Noisy	SNP $h^2 > 5\%$, PRS(r^2) > 0.5%	Inverse Transformed Data	(167,97), [75,16]

Fig. 7: Secondary QC

Trait	All Samples		European Samples		Sanjak et al					
	Linear Parameter (β)	Quadratic Parameter (γ)	Linear Parameter (β)	Quadratic Parameter (γ)	Aggregate Result		Males		Females	
					β	γ	β	γ	β	γ
Bmi	0.05* (positive)	-0.028* (stabilizing)	0.058* (positive)	-0.048* (stabilizing)	positive	stabilizing	0.072*	-0.0327*	0.0601*	-0.0327*
Basal Metabolic Rate	0.045* (positive)	-0.019* (stabilizing)	0.083* (positive)	-0.036* (stabilizing)	positive	stabilizing	0.0747*	-0.03*	0.0405*	-0.03*
Birth Weight	0.011* (positive)	-0.007* (stabilizing)	0.019* (positive)	-0.002	positive	stabilizing	0.0208*	-0.0016	0.0047	-0.0016
Body Fat Percentage	0.019* (positive)	-0.019* (stabilizing)	0.01* (positive)	-0.032* (stabilizing)	positive	stabilizing	0.0198*	-0.0273*	0.0188*	-0.0273*
Fathers Age	-0.074* (negative)	0.005	-0.074* (negative)	0.005	negative	diverging	-0.058*	0.0004	-0.0744*	0.0004
Fluid Intelligence Score	-0.029* (negative)	0.008* (diverging)	-0.02	0.007	positive/negative	diverging	0.0139	0.0009	-0.0473	0.0009
Forced Expiratory Vol	0.016* (positive)	-0.007* (stabilizing)	0.023* (positive)	-0.007* (stabilizing)	positive	stabilizing	0.0335*	-0.0109*	0.0126	-0.0109*
Forced Vital Capacity Fvc	0.018* (positive)	-0.009* (stabilizing)	0.023* (positive)	-0.011* (stabilizing)	positive	stabilizing	0.0323*	-0.0141*	0.0165*	-0.0141*
Hand Grip Strength Left	0.044* (positive)	-0.002	0.044* (positive)	-0.001	positive	stabilizing	0.0728*	-0.0023	0.0124*	-0.0023
Hand Grip Strength Right	0.038* (positive)	-0.001	0.038* (positive)	-0.001	positive	stabilizing	0.0656*	-0.0026	0.0047*	-0.0026
Heel BMD	0.006	-0.013* (stabilizing)	0.006	-0.013* (stabilizing)	positive/negative	stabilizing	0.0325*	-0.0094*	-0.0105	-0.0094*
Heel BMD T Score	0.006	-0.013* (stabilizing)	0.006	-0.013* (stabilizing)	positive/negative	stabilizing	0.0314*	-0.0092*	-0.0104	-0.0092*
Heel Qui	0.006	-0.013* (stabilizing)	0.006	-0.013* (stabilizing)	positive/negative	stabilizing	0.0314*	-0.0092*	-0.0104	-0.0092*
Height	-0.006* (negative)	-0.014* (stabilizing)	-0.006* (negative)	-0.014* (stabilizing)	positive/negative	stabilizing	0.0229*	-0.0152*	-0.0283*	-0.0152*
Highest Edu	-0.068* (negative)	0.013* (diverging)	-0.068* (negative)	0.013* (diverging)	positive/negative	diverging	0.0086	0.0313*	-0.0612	0.0313*
Hip Circumference	0.034* (positive)	-0.022* (stabilizing)	0.034* (positive)	-0.022* (stabilizing)	positive	stabilizing	0.0505*	-0.0278*	0.0357*	-0.0278*
Maximum Digits Remembered	-0.004	-0.01* (stabilizing)	0.016	-0.01* (stabilizing)	negative	stabilizing	-0.0011	-0.0014	-0.0248	-0.0014
Time to Identify Matches	-0.021* (negative)	-0.005* (stabilizing)	-0.042* (negative)	-0.007* (stabilizing)	negative	stabilizing	-0.0434*	-0.0002	-0.0101*	-0.0002
Neuroticism Score	-0.018* (negative)	-0.01* (stabilizing)	-0.038* (negative)	-0.012* (stabilizing)	negative	stabilizing	-0.0318*	-0.0067	-0.0003	-0.0067
Pulse Rate	-0.022* (negative)	-0.011* (stabilizing)	-0.022* (negative)	-0.011* (stabilizing)	negative	stabilizing	-0.0316*	-0.0062*	-0.0136*	-0.0062*
Waist Circumference	0.053* (positive)	-0.029* (stabilizing)	0.053* (positive)	-0.029* (stabilizing)	positive	stabilizing	0.0459*	-0.0331*	0.0649*	-0.0331*
Weight	0.047* (positive)	-0.028* (stabilizing)	0.07* (positive)	-0.047* (stabilizing)	positive	stabilizing	0.0761*	-0.0343*	0.0439*	-0.0343*

Fig. 8: **Comparison of Selection Estimates**